

The nonsolvable of order 15 360

Cross-checks by the Zphi and supplement method

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Extensions of Alt(5) by groups of order 256

#grps	$\mathcal{O}_K$	total
56 092	1 027 380	1 083 472

Extensions of SL(2,5) by groups of order 128

$\mathcal{O}_Z$	$\mathcal{O}_{Z,K}$	total
337 956	6 856 498	7 194 454

Extensions of perfect groups of order 1920

P	total
$P_1 = \text{PerfectGroup}(1920, 1)$	261
$P_2 = \text{PerfectGroup}(1920, 2)$	261
$P_3 = \text{PerfectGroup}(1920, 3)$	196
$P_4 = \text{PerfectGroup}(1920, 4)$	34
$P_5 = \text{PerfectGroup}(1920, 5)$	63
$P_6 = \text{PerfectGroup}(1920, 6)$	63
$P_7 = \text{PerfectGroup}(1920, 7)$	63
Total	941

Extensions of perfect groups of order 3840

P	total
$P_1 = \text{PerfectGroup}(3840, 1)$	40
$P_2 = \text{PerfectGroup}(3840, 2)$	40
$P_3 = \text{PerfectGroup}(3840, 3)$	60
$P_4 = \text{PerfectGroup}(3840, 4)$	131
$P_5 = \text{PerfectGroup}(3840, 5)$	240
$P_6 = \text{PerfectGroup}(3840, 6)$	22
$P_7 = \text{PerfectGroup}(3840, 7)$	68
Total	601

Order 15 360

Extensions of perfect groups of order 7680

P	total
$P_1 = \text{PerfectGroup}(7680, 1)$	11
$P_2 = \text{PerfectGroup}(7680, 2)$	10
$P_3 = \text{PerfectGroup}(7680, 3)$	10
$P_4 = \text{PerfectGroup}(7680, 4)$	16
$P_5 = \text{PerfectGroup}(7680, 5)$	10
Total	57
Total PerfectGroups(15360)	7

Total non solvable groups of order 15360

family	total
Extensions of Alt(5) by groups of order 256	1 083 472
Extensions of SL(2,5) by groups of order 128	7 194 454
Extensions of perfect groups of order 1920	941
Extensions of perfect groups of order 3840	601
Extensions of perfect groups of order 7680	57
Total PerfectGroups(15360)	7
Global total	8 279 532